

INDEX

S.NO	DATE	TITLE
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EXP NO: 1 DATE:	AZURE DEVOPS ENVIRONMENT SETUP
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AIM:-

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

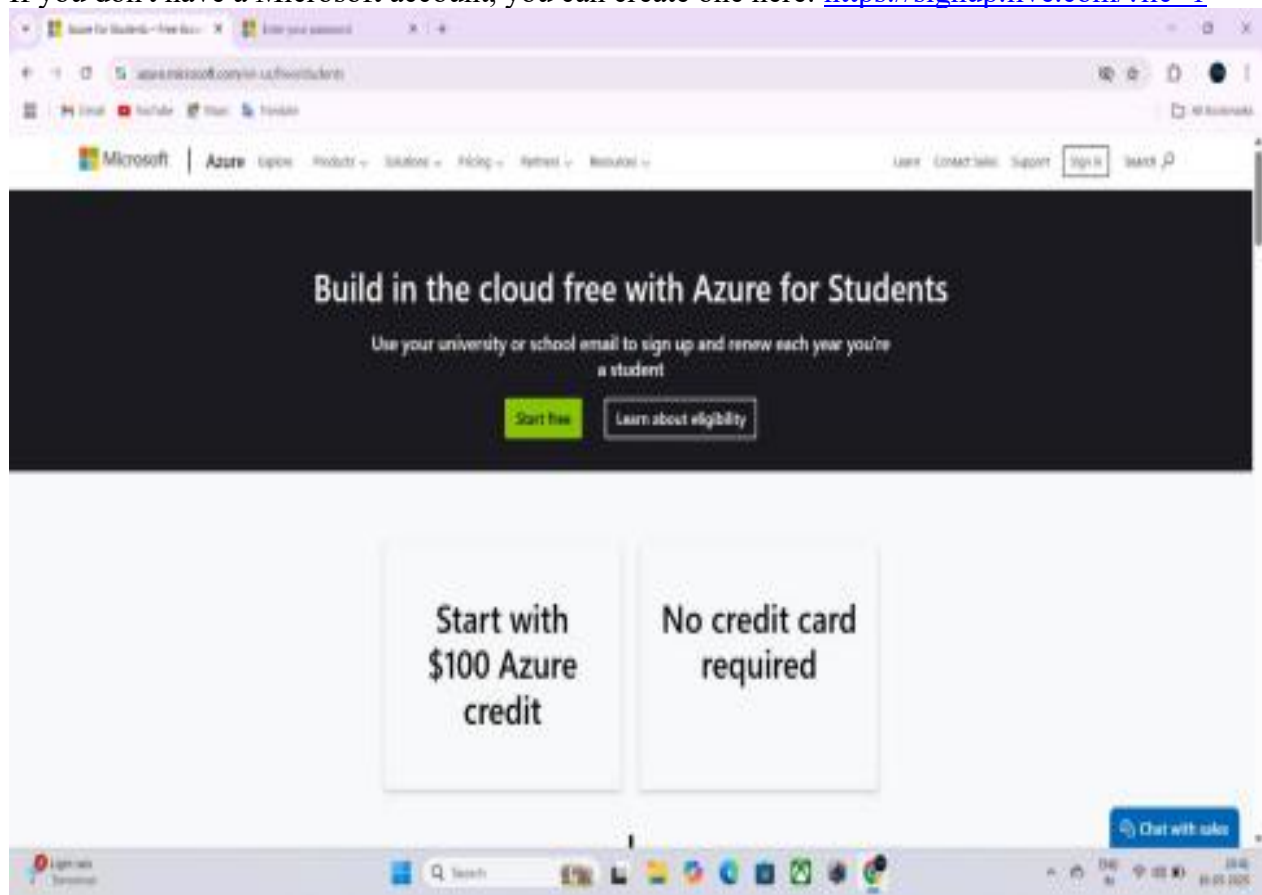
INSTALLATION:-

1. Open your web browser and go to the Azure website:

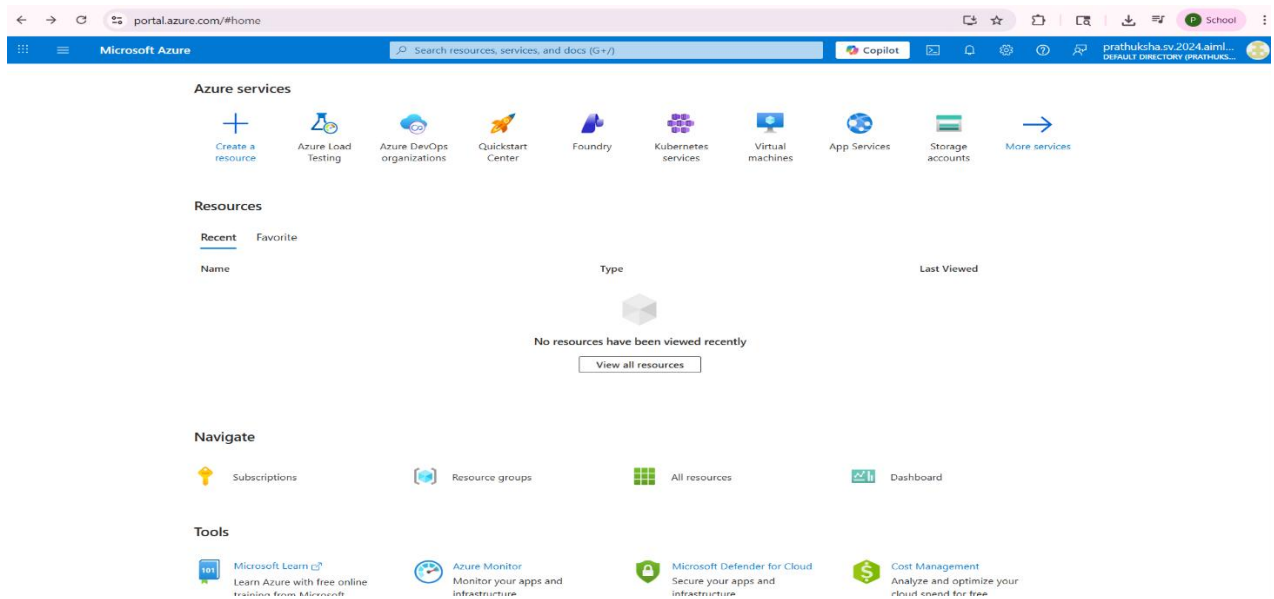
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

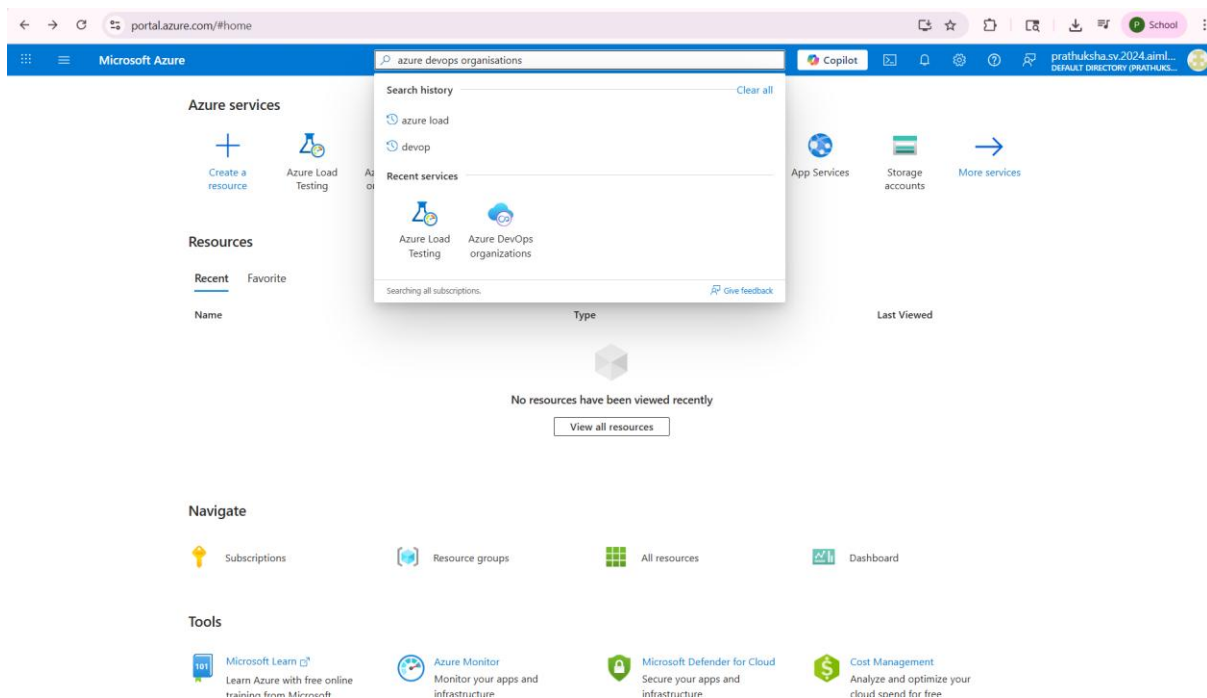
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



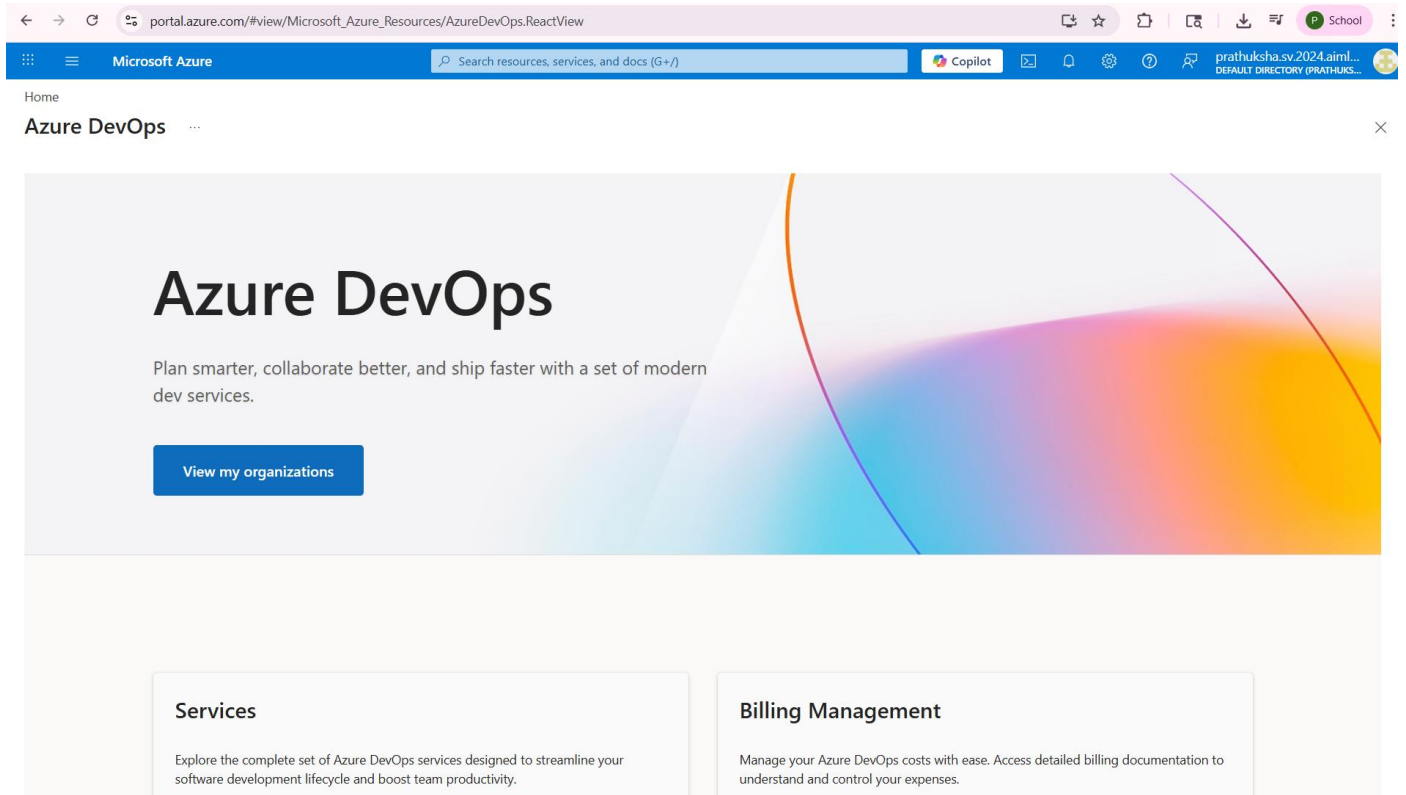
2.Azure home page



3.Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



RESULT:

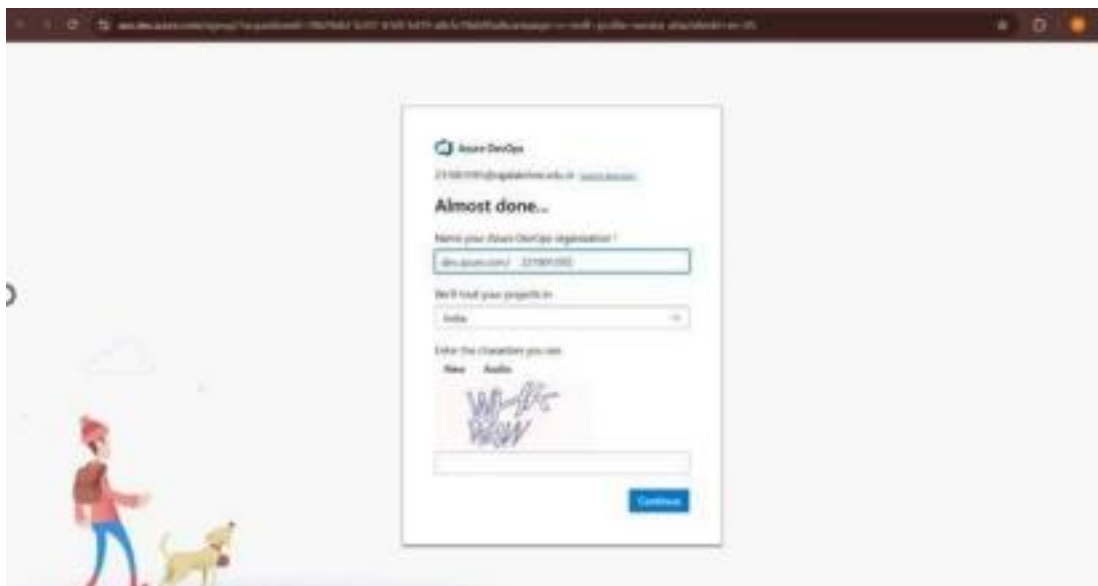
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2 DATE:	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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AIM:-

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Create new project



Project name *

Expense Tracker

Description

Visibility

 **Private**

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ **Advanced**

Version control ⓘ

Git



Work item process ⓘ

Agile



Cancel

Create

← → ↺ aex.dev.azure.com/me?mkt=en-US ☆ 📄 📄 📄 📄 📄 School ⋮

Microsoft prathuksha SV Sign out



prathuksha SV [Edit profile](#)

prathuksha.sv.2024.aiiml@rajalakshmi.edu.in

Microsoft account

India

prathuksha.sv.2024.aiiml@rajalakshmi.edu.in

Visual Studio Dev Essentials

Get everything you need to build and deploy your app on any platform.

[Use your benefits](#)

Azure DevOps Organizations

[Create new organization](#)

dev.azure.com/prathukshasv2024aiiml (Owner)

Projects

- Expense Tracker System
- Online food waste donation system
- [New project](#)

Actions

- [Open in Visual Studio](#)

4. Project dashboard

← → ↺ dev.azure.com/prathukshasv2024aiiml/Expense%20Tracker%20System ☆ 📄 📄 📄 📄 📄 School ⋮

Azure DevOps prathukshasv2024aiiml / Expense Tracker System / Overview / Summary 🔍 Search ⋮ 🔒 ⚙️ PS

Expense Tracker System +

- Overview
- Summary**
- Dashboards
- Wiki
- Boards
- Repos
- Pipelines
- Test Plans
- Artifacts

ES Expense Tracker System Private [Invite](#) ☆

About this project

Help others to get on board!

Describe your project and make it easier for other people to understand it.

[+ Add Project Description](#)

Project stats

Period: Last 7 days

Boards

- 0 Work items created
- 0 Work items completed

Repos

- 0 Pull requests opened
- 7 Commits by 1 authors

Members 3

- PS
- MS
- SV

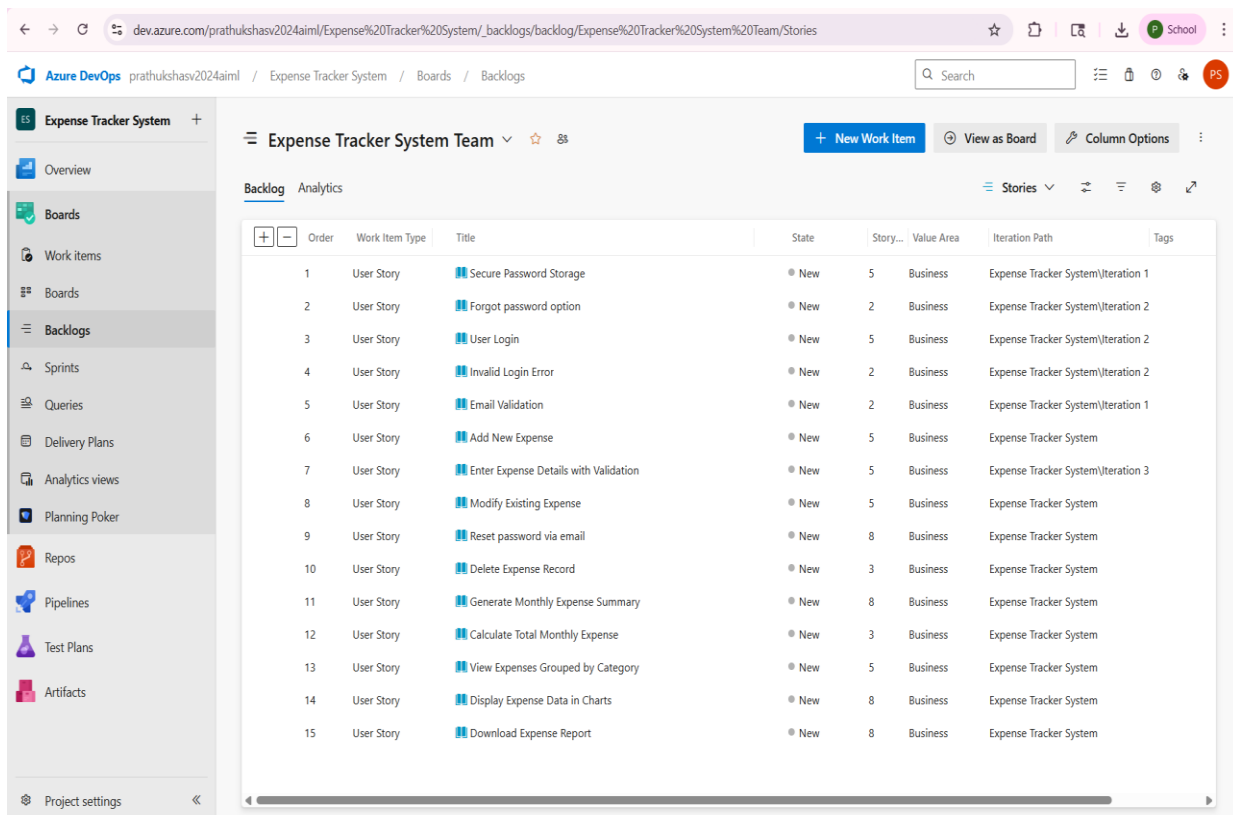
Print outline

[https://dev.azure.com/prathukshasv2024aiiml/Expense Tracker System](https://dev.azure.com/prathukshasv2024aiiml/Expense%20Tracker%20System)

5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



The screenshot shows the Azure DevOps interface for the 'Expense Tracker System' project. The left-hand navigation menu is visible, with 'Boards' selected. The main area displays the 'Backlog' view for the 'Expense Tracker System Team'. The backlog contains 15 user stories, each with an order, title, state, story number, value area, iteration path, and tags.

Order	Work Item Type	Title	State	Story...	Value Area	Iteration Path	Tags
1	User Story	Secure Password Storage	New	5	Business	Expense Tracker System\Iteration 1	
2	User Story	Forgot password option	New	2	Business	Expense Tracker System\Iteration 2	
3	User Story	User Login	New	5	Business	Expense Tracker System\Iteration 2	
4	User Story	Invalid Login Error	New	2	Business	Expense Tracker System\Iteration 2	
5	User Story	Email Validation	New	2	Business	Expense Tracker System\Iteration 1	
6	User Story	Add New Expense	New	5	Business	Expense Tracker System	
7	User Story	Enter Expense Details with Validation	New	5	Business	Expense Tracker System\Iteration 3	
8	User Story	Modify Existing Expense	New	5	Business	Expense Tracker System	
9	User Story	Reset password via email	New	8	Business	Expense Tracker System	
10	User Story	Delete Expense Record	New	3	Business	Expense Tracker System	
11	User Story	Generate Monthly Expense Summary	New	8	Business	Expense Tracker System	
12	User Story	Calculate Total Monthly Expense	New	3	Business	Expense Tracker System	
13	User Story	View Expenses Grouped by Category	New	5	Business	Expense Tracker System	
14	User Story	Display Expense Data in Charts	New	8	Business	Expense Tracker System	
15	User Story	Download Expense Report	New	8	Business	Expense Tracker System	

dev.azure.com/prathukshasv2024aiml/Expense%20Tracker%20System/_backlogs/backlog/Expense%20Tracker%20System%20Team/Stories

Azure DevOps prathukshasv2024aiml / Expense Tracker System / Boards / Backlogs

Expense Tracker System Team

Backlog Analytics

	Order	Work Item Type	Title	State	Story...	Value Area
1	User Story	Secure Password Storage	New	5	Business	
2	User Story	Forgot password option	New	2	Business	
3	User Story	User Login	New	5	Business	
4	User Story	Invalid Login Error	New	2	Business	
5	User Story	Email Validation	New	2	Business	
6	User Story	Add New Expense	New	5	Business	
7	User Story	Enter Expense Details with Validation	New	5	Business	
8	User Story	Modify Existing Expense	New	5	Business	
9	User Story	Reset password via email	New	8	Business	
10	User Story	Delete Expense Record	New	3	Business	
11	User Story	Generate Monthly Expense Summary	New	8	Business	
12	User Story	Calculate Total Monthly Expense	New	3	Business	
13	User Story	View Expenses Grouped by Category	New	5	Business	
14	User Story	Display Expense Data in Charts	New	8	Business	
15	User Story	Download Expense Report	New	8	Business	

prathuksha SV
prathuksha.sv.2024.aiml@raja...
My Microsoft account
Switch directory

Expense Tracker System\Iteration 2
Expense Tracker System\Iteration 2
Expense Tracker System\Iteration 1
Expense Tracker System
Expense Tracker System\Iteration 3
Expense Tracker System
Expense Tracker System
Expense Tracker System
Expense Tracker System
Expense Tracker System
Expense Tracker System
Expense Tracker System

Result

Successfully created an Azure DevOps project with user story management and agile work flow setup.

2116241501149

CS23432

EXP NO: 3
DATE:

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot displays the Azure DevOps interface for the 'Expense Tracker System Team'. The main view is the 'Backlog' tab, showing a list of work items organized into three epics:

- Epic 1: User Authentication**
 - Feature: User Registration
 - User Story: Secure Password Storage
 - User Story: Email Validation
 - Feature: User login
 - User Story: User Login
 - User Story: Invalid Login Error
- Epic 2: Expense Management**
 - Feature: Password Management
 - User Story: Forgot password option
 - User Story: Reset password via email
- Epic 3: Reports and Analytics**
 - Feature: Add Expense
 - Feature: Edit Expense
 - Feature: Delete Expense
 - Feature: Monthly Report

The right sidebar shows the 'Planning' view, which includes a sprint plan for 'Expense Tracker System Team backlog'. The sprint plan shows three iterations:

- Iteration 1** (2/24/2026 - 3/1/2026): Planned Effort: 0, 4 working days. Contains 2 items.
- Iteration 2**: Contains 1 item.
- Iteration 3**: Planned Effort: 0. Contains 1 item.

A '+ New Sprint' button is visible at the bottom of the sprint plan.

1.Fill in Epics

dev.azure.com/prathukhasv2024aiml/Expense%20Tracker%20System/_workitems/create/Epic

Azure DevOps prathukhasv2024aiml / Expense Tracker System / Boards / Work items

Expense Tracker System

Work Items Back to Work Items

NEW EPIC *

User Autentication

No one selected 0 Comments Add Tag Save

State New Area Expense Tracker System Reason New Iteration Expense Tracker System/Iteration 3

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request. switch to Markdown editor

Planning

Priority 2 Risk Effort Business Value Time Criticality Start Date Select a date... Target Date Select a date...

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link Add an existing work item as a parent

Classification

Value area Business

2.Fill in Features

dev.azure.com/prathukhasv2024aiml/Expense%20Tracker%20System/_workitems/create/Feature

Azure DevOps prathukhasv2024aiml / Expense Tracker System / Boards / Work items

Expense Tracker System

Work Items Back to Work Items

NEW FEATURE *

User Registration

No one selected 0 Comments Add Tag Save

State New Area Expense Tracker System Reason New Iteration Expense Tracker System/Iteration 3

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request. switch to Markdown editor

Planning

Priority 2 Risk Effort Business Value Time Criticality Start Date Select a date... Target Date Select a date...

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link Add an existing work item as a parent

Classification

Value area Business

3.Fill in User Story Details

The screenshot shows the 'New User Story' form in Azure DevOps. The browser address bar shows the URL: `dev.azure.com/prathukhasv2024aiml/ExpenseTracker%20System/_workitems/create/User%20Story`. The left sidebar contains a navigation menu with options: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Artifacts. The main content area is titled 'NEW USER STORY *' and 'Secure password storage'. It includes a 'Save' button and a 'Details' tab. The form is divided into several sections: Description (with the text 'As a user, I want my password to be stored securely so that my account information is protected.'), Acceptance Criteria (with a 'Click to add Acceptance Criteria.' prompt and a 'switch to Markdown editor' link), Discussion (with a 'Add a comment...' prompt), Planning (with fields for Story Points, Priority (set to 2), and Risk), Classification (with fields for Value area and Business), Deployment (with instructions on tracking releases), Development (with an 'Add link' prompt and instructions on linking to Azure Repos), and Related Work (with an 'Add link' prompt and a link to 'Add an existing work item as a parent').

RESULT:-

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4
DATE:

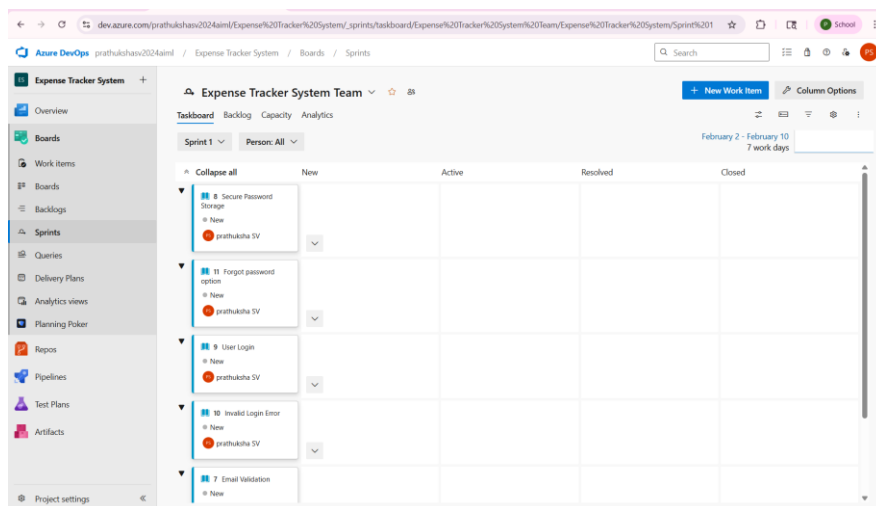
SPRINT PLANNING

AIM:-

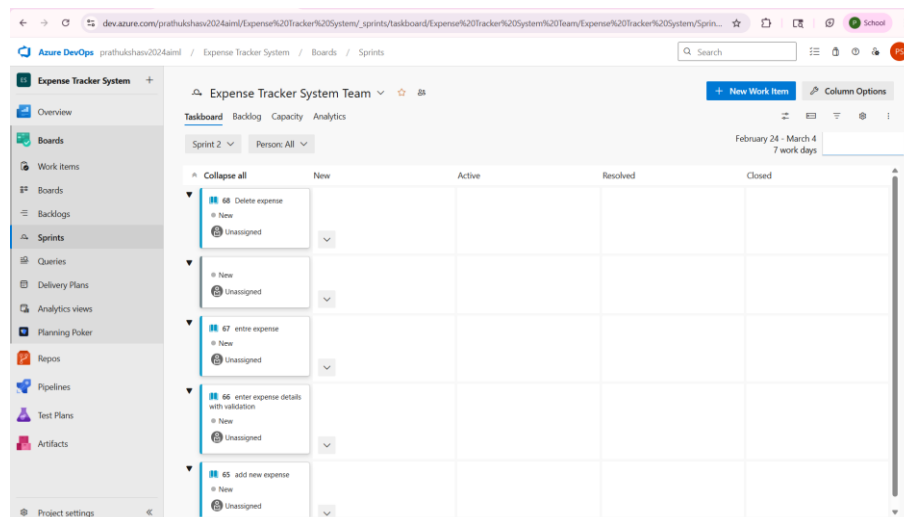
To assign user story to specific sprint for the Student Management System Project.

Sprint Planning:-

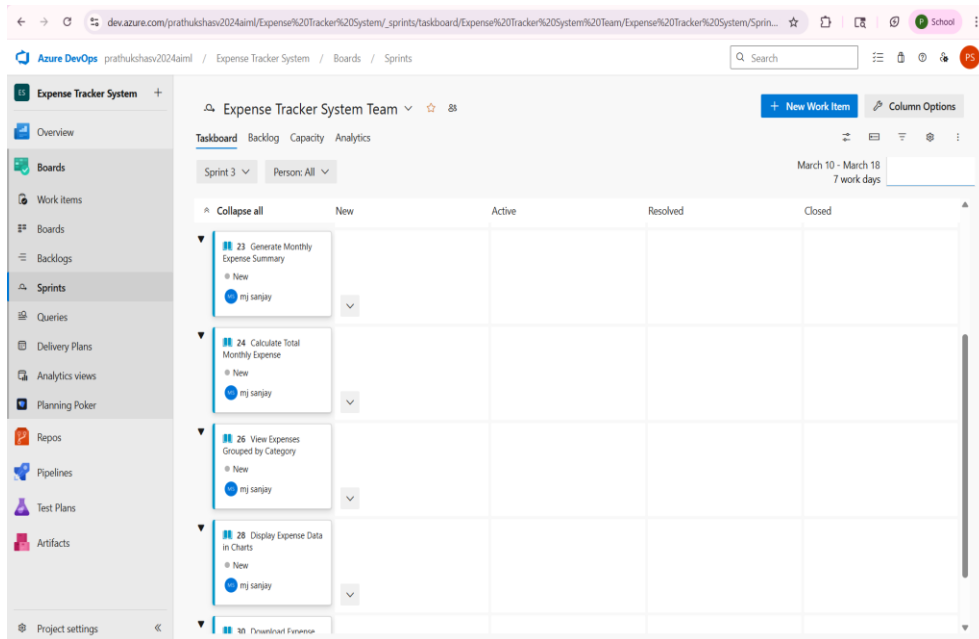
Sprint 1



Sprint 2



Sprint 3



RESULT:-

The Sprints are created for the Student Management System.

EXP NO: 5
DATE:

POKER ESTIMATION

AIM:-

Create Poker Estimation for the user stories -Student management System.

Poker Estimation:-

The screenshot displays the Azure DevOps interface for a project named 'Expense Tracker System'. The left sidebar shows the navigation menu with options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Artifacts. The main area shows a user story titled '11 Forgot password option' created by 'prathuksha SV'. The story is in the 'New' state and is associated with the 'Expense Tracker System' area and 'Expense Tracker System/Sprint 1' iteration. The description states: 'As a user, I want a "Forgot Password" option so that I can reset my password if I forget it.' The acceptance criteria are: 'Forgot password link must be available on login page', 'User must enter registered email', and 'System must verify email existence'. The planning section shows 'Story Points' as 2, 'Priority' as 2, and 'Risk' as 2. The classification is 'Business'. The planning poker section shows a vote of 3, with buttons for 0, 1, 2, 3, 5, 8, and 13. The deployment section provides instructions on how to track releases and turn on deployment status reporting. The development section includes an 'Add link' button to link an Azure Repos commit, pull request, or branch. The related work section shows a dropdown for 'Add link' and a 'Parent' link.

RESULT:-

The Estimation/Story Points is created for the project using Poker Estimation.

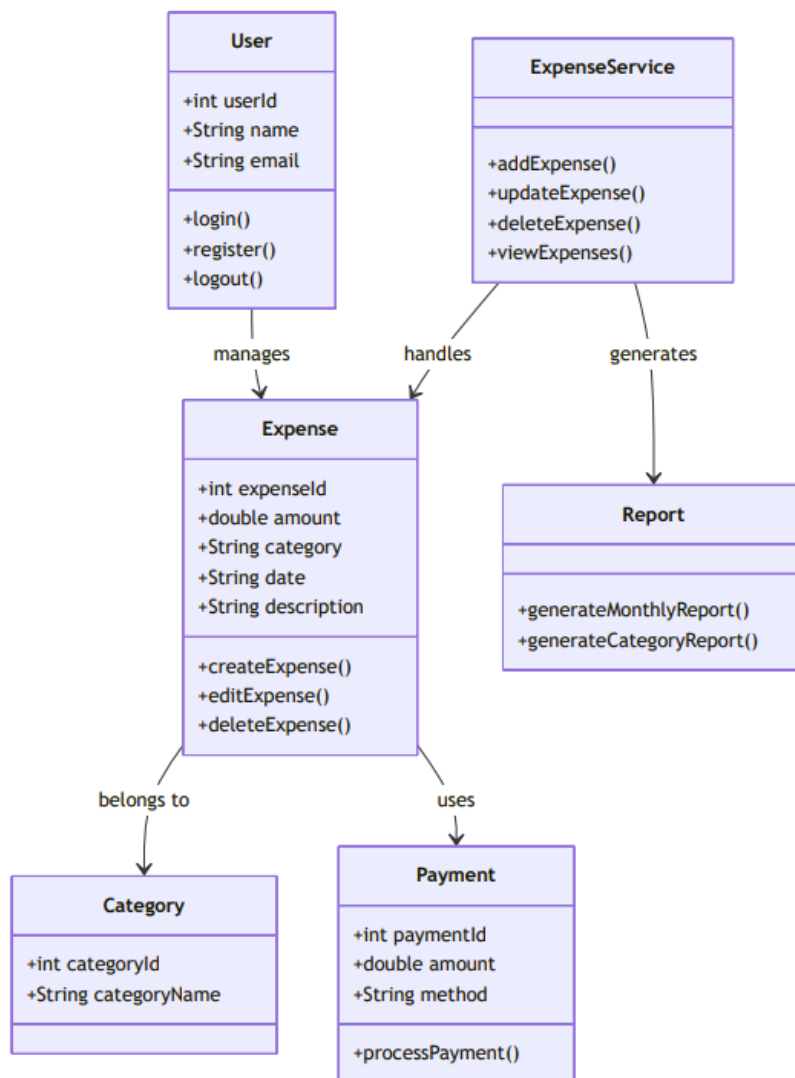
EXP NO: 6
DATE:

**DESIGNING CLASS AND SEQUENCE DIAGRAMS
FOR PROJECT ARCHITECTURE**

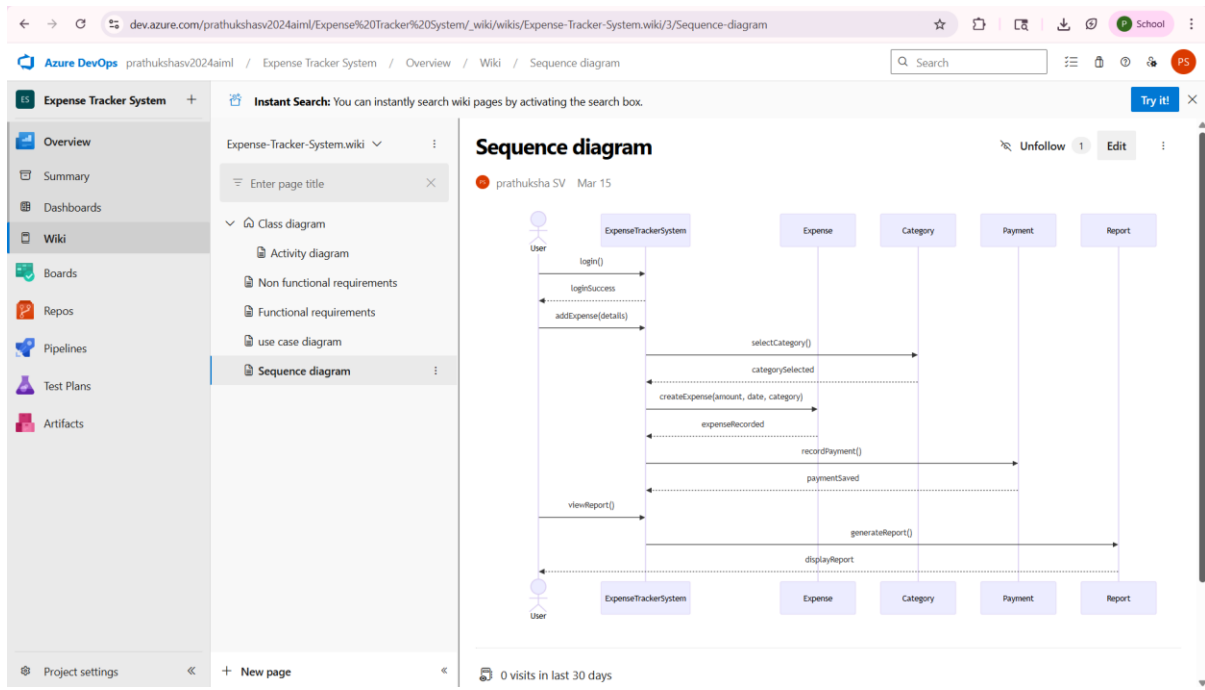
AIM:-

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. CLASS DIAGRAM:-



6B. SEQUENCE DIAGRAM:-



Result:

The Class Diagram and Sequence Diagram is designed Successfully for given project.

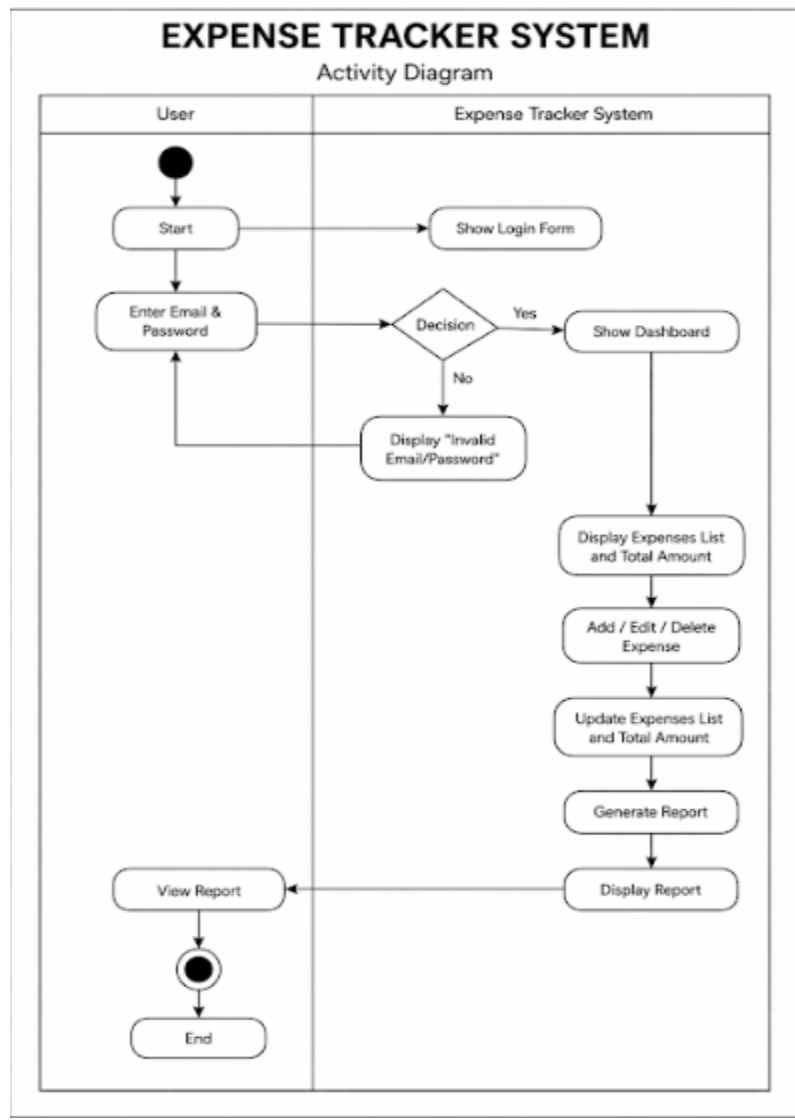
EXP NO: 7
DATE:

**DESIGNING USE CASE DIAGRAMS AND
ACTIVITY DIAGRAM FOR PROJECT
ARCHITECTURE**

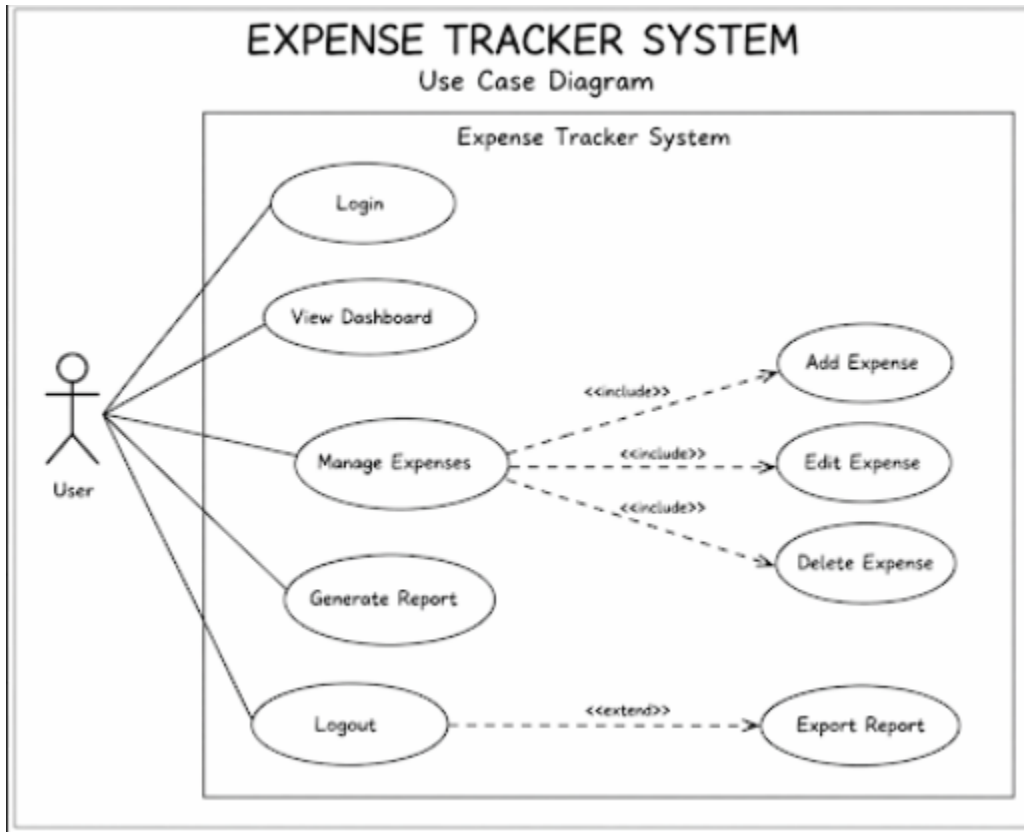
AIM:-

To Design an activity Diagram and use case Diagram for the given Project.

7A. ACTIVITY DIAGRAM:-



7B. USE CASE DIAGRAM:-



RESULT:-

Thus activity and use case diagram has been designed successfully for Student Management System

EXP NO: 8
DATE:

TESTING – TEST PLANS AND TEST CASES

AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case :

Test Case Design Procedure :

1. Understand Core Features of the Application

- User Signup & Login
- Student Enrolment in Courses
- Assignment Submission & Evaluation
- Viewing Grades and Attendance
- Admin Role Management and Report Generation

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, submitting an assignment, viewing results).

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions.
- **Example:** Student registers and logs in, enrolls in a course, submits assignment, and views grades.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- **Example:** Login with wrong credentials, submission without attachment, unauthorized access to admin panel.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identified (e.g., TC01 – Student Login Success, TC12 – Invalid Assignment Submission).
- Facilitates easy mapping to features and tracking in Azure DevOps

7. Separate Test Suite

Grouped by functionality such as:

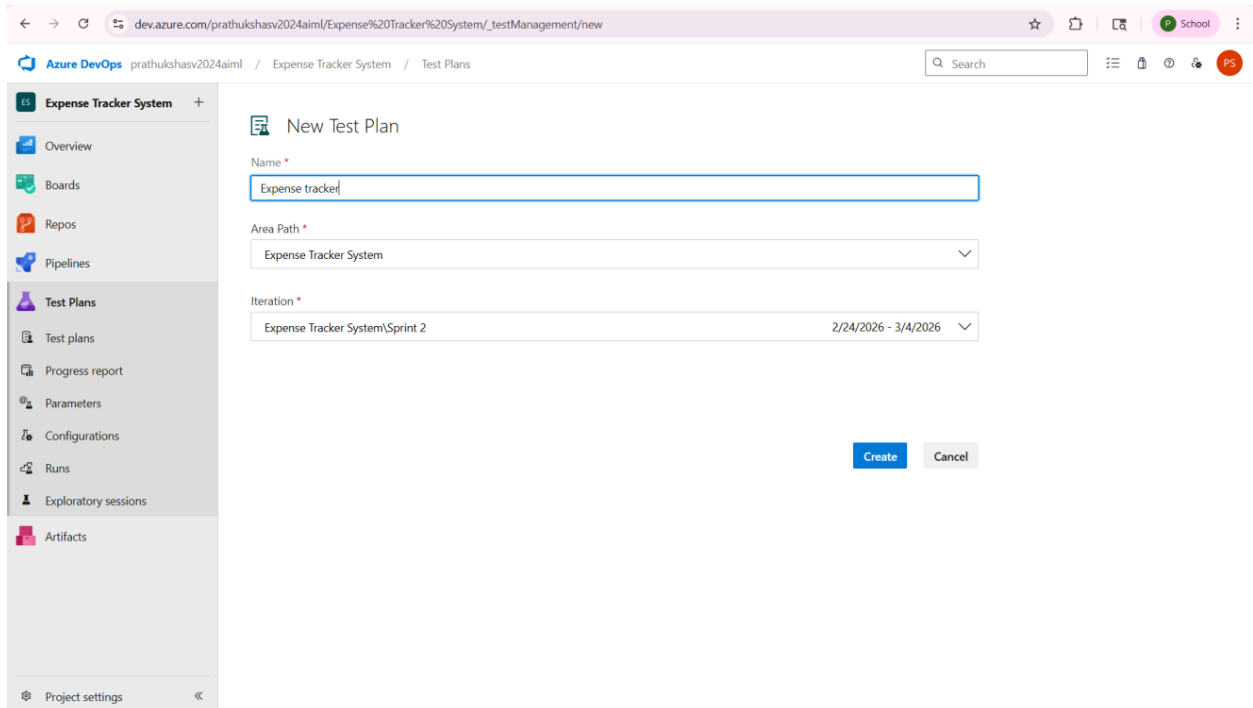
- o Login and Registration
- o Course Enrolment
- o Assignment Submission
- o Report Generation
- o Admin Functions

· Improves organization and enables focused execution in Azure DevOps.

8. Prioritize and Review

· High-priority assigned to critical workflows like login, course access, and grading. · Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

1. New test plan

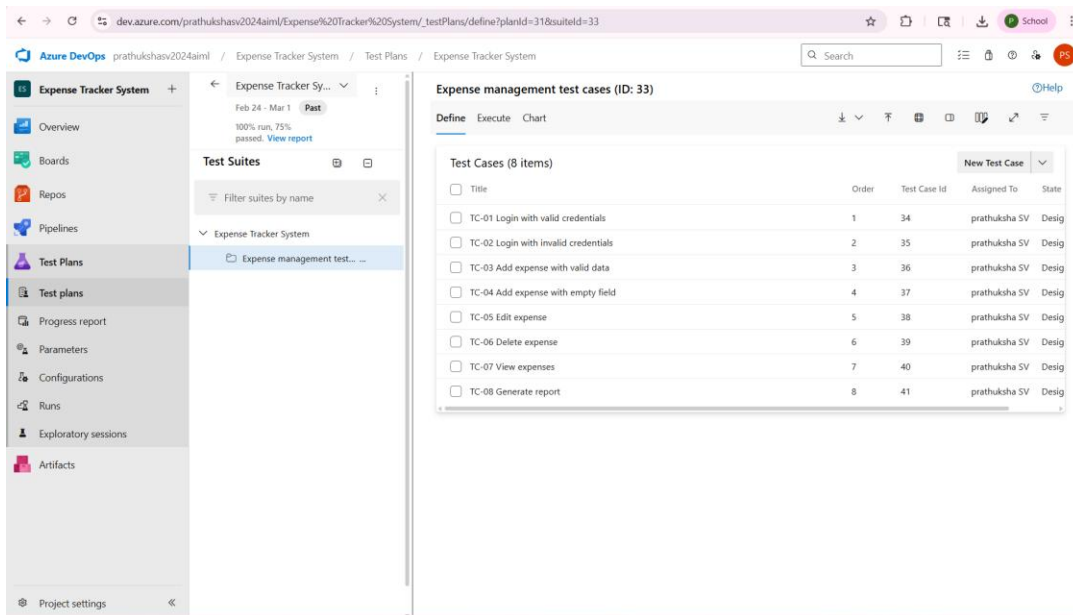


The screenshot shows the 'New Test Plan' form in the Azure DevOps web interface. The browser address bar shows the URL: `dev.azure.com/prathukhasv2024aiml/Expense%20Tracker%20System/_testManagement/new`. The left sidebar contains a navigation menu with the following items: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main content area is titled 'New Test Plan' and contains the following fields:

- Name ***: A text input field containing 'Expense tracker'.
- Area Path ***: A dropdown menu showing 'Expense Tracker System'.
- Iteration ***: A dropdown menu showing 'Expense Tracker System\Sprint 2' with a date range of '2/24/2026 - 3/4/2026'.

At the bottom right of the form, there are two buttons: 'Create' (in blue) and 'Cancel' (in grey).

2. Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Management System– Test Plans

USER STORIES:-

- As a user, I want to sign up and log in securely so that I can access my expense records. **(ID: 101)**
- As a user, I want to add expenses so that I can track my daily spending. **(ID: 102)**
- As a user, I want to edit or delete expenses so that I can correct or remove entries. **(ID: 103)**
- As a user, I want to view all my expenses in one place so that I can monitor my spending. **(ID: 104)**
- As a user, I want to generate reports so that I can analyse my expenses. **(ID: 105)**

Test Suite: TS01 – Expense Tracker System (ID: 201)

TC01 – Login with Valid Credentials

- **Steps:**
 1. Open login page
 2. Enter valid email & password
 3. Click “Login”
- **Expected Result:**
User is logged in and dashboard is displayed
- **Type:** Happy Path

TC02 – Login with Invalid Credentials

- **Steps:**
 1. Enter invalid email or password
 2. Click “Login”
- **Expected Result:**
Error message “Invalid credentials” is displayed
- **Type:** Error Path

TC03 – Add Expense with Valid Data

- **Steps:**
 1. Login → Click “Add Expense”
 2. Enter amount, category, date
 3. Click “Save”
- **Expected Result:**
Expense is added successfully and shown in list
- **Type:** Happy Path

TC04 – Add Expense with Empty Fields

- **Steps:**
 1. Click “Add Expense”
 2. Leave fields empty
 3. Click “Save”
- **Expected Result:**
Error message “Fields cannot be empty”
- **Type:** Error Path

TC05 – Edit Expense

- **Steps:**
 1. Select an existing expense
 2. Click “Edit”
 3. Modify details and save
- **Expected Result:**
Expense details are updated successfully
- **Type:** Happy Path

TC06 – Delete Expense

- **Steps:**
 1. Select an expense
 2. Click “Delete”
- **Expected Result:**
Expense is removed from list
- **Type:** Happy Path

TC07 – View Expenses

- **Steps:**
 1. Login
 2. Open dashboard
- **Expected Result:**
All expenses are displayed with total amount
- **Type:** Happy Path

TC08 – Generate Report

- **Steps:**
 1. Click “Generate Report”
 2. Select date range
 3. Click generate
- **Expected Result:**
Report is generated with expense summary
- **Type:** Happy Path

TC09 – Generate Report with No Data

- **Steps:**
 1. Select date range with no expenses
 2. Click generate
- **Expected Result:**
Message “No data available” is shown
- **Type:** Error Path

TC10 – Logout

- **Steps:**
 1. Click logout button
- **Expected Result:**
User is logged out and redirected to login page
- **Type:** Happy Path

3. Test Cases

dev.azure.com/prathukshasv2024aiml/Expense%20Tracker%20System/_testManagement/new

Azure DevOps prathukshasv2024aiml / Expense Tracker System / Test Plans

Expense Tracker System +

- Overview
- Boards
- Repos
- Pipelines
- Test Plans**
 - Test plans
 - Progress report
 - Parameters
 - Configurations
 - Runs
 - Exploratory sessions
- Artifacts

Project settings <<

New Test Plan

Name *
Expense tracker

Area Path *
Expense Tracker System

Iteration *
Expense Tracker System\Sprint 2 2/24/2026 - 3/4/2026

Create Cancel

dev.azure.com/prathukshasv2024aiml/Expense%20Tracker%20System/_testPlans/define?planId=31&suiteId=33

Azure DevOps prathukshasv2024aiml / Expense Tracker System / Test Plans / Expense Tracker System

Expense Tracker Sy...
Feb 24 - Mar 1
100% run, 75% passed. View report

Test Suites

Filter suites by name

Expense Tracker System

Expense management test... ..

Expense management test cases (ID: 33)

Define Execute Chart

Test Cases (8 items)

	Order	Test Case Id	Assigned To	State
☐ Title				
☐ TC-01 Login with valid credentials	1	34	prathuksha SV	Desig
☐ TC-02 Login with invalid credentials	2	35	prathuksha SV	Desig
☐ TC-03 Add expense with valid data	3	36	prathuksha SV	Desig
☐ TC-04 Add expense with empty field	4	37	prathuksha SV	Desig
☐ TC-05 Edit expense	5	38	prathuksha SV	Desig
☐ TC-06 Delete expense	6	39	prathuksha SV	Desig
☐ TC-07 View expenses	7	40	prathuksha SV	Desig
☐ TC-08 Generate report	8	41	prathuksha SV	Desig

dev.azure.com/prathukshav2024aiml/Expense%20Tracker%20System/_testPlans/define?planId=31&suiteId=33

TEST CASE 34

34 TC-01 Login with valid credentials

prathuksha SV 0 Comments Add Tag

State: Design Area: Expense Tracker System Reason: New Iteration: Expense Tracker System\Sprint 1

Updated by prathuksha SV: 6 Apr

Steps Summary Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	open applications	login successful	
2.	Enter valid username and password		
3.	click login		

Click or type here to add a step

Recent test results

Passed by prathuksha SV with Windows 10 Completed 7:40 PM, 4/3/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

[Add an existing work item as a parent](#)

dev.azure.com/prathukshav2024aiml/Expense%20Tracker%20System/_testPlans/define?planId=31&suiteId=33

TEST CASE 36

36 TC-03 Add expense with valid data

prathuksha SV 0 Comments Add Tag

State: Design Area: Expense Tracker System Reason: New Iteration: Expense Tracker System\Sprint 1

Updated by prathuksha SV: 6 Apr

Steps Summary Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	login	Expense added successfully	
2.	enter expense details		
3.	click save		

Click or type here to add a step

Recent test results

Passed by prathuksha SV with Windows 10 Completed 7:41 PM, 4/3/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

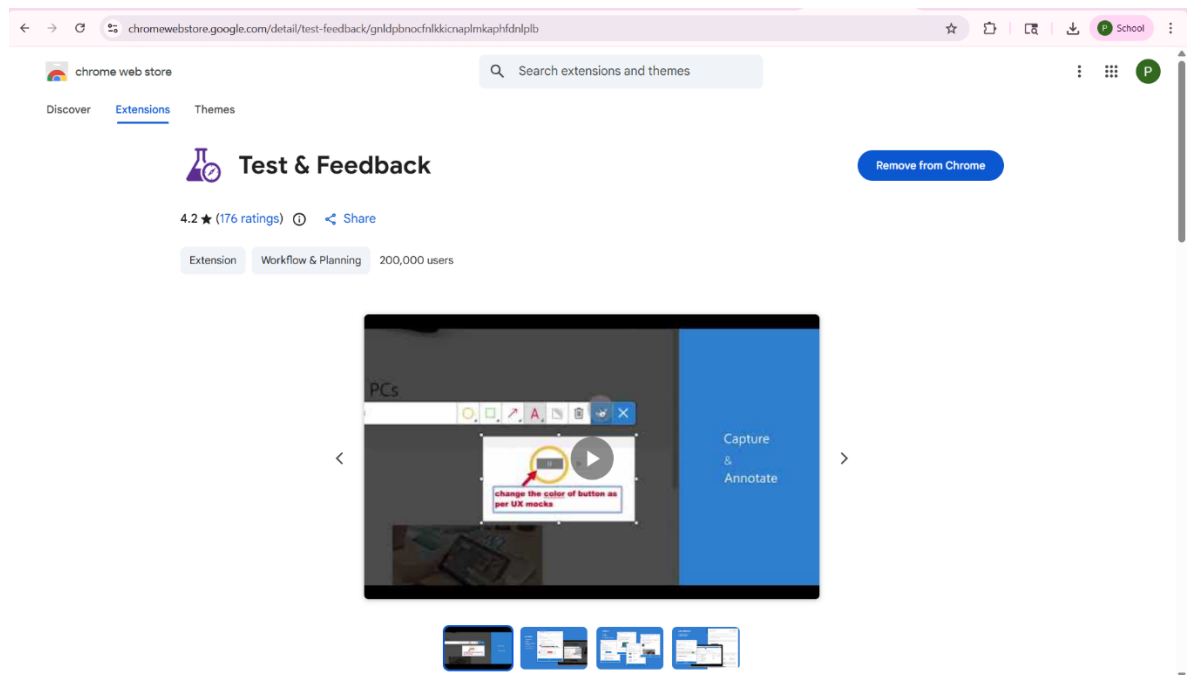
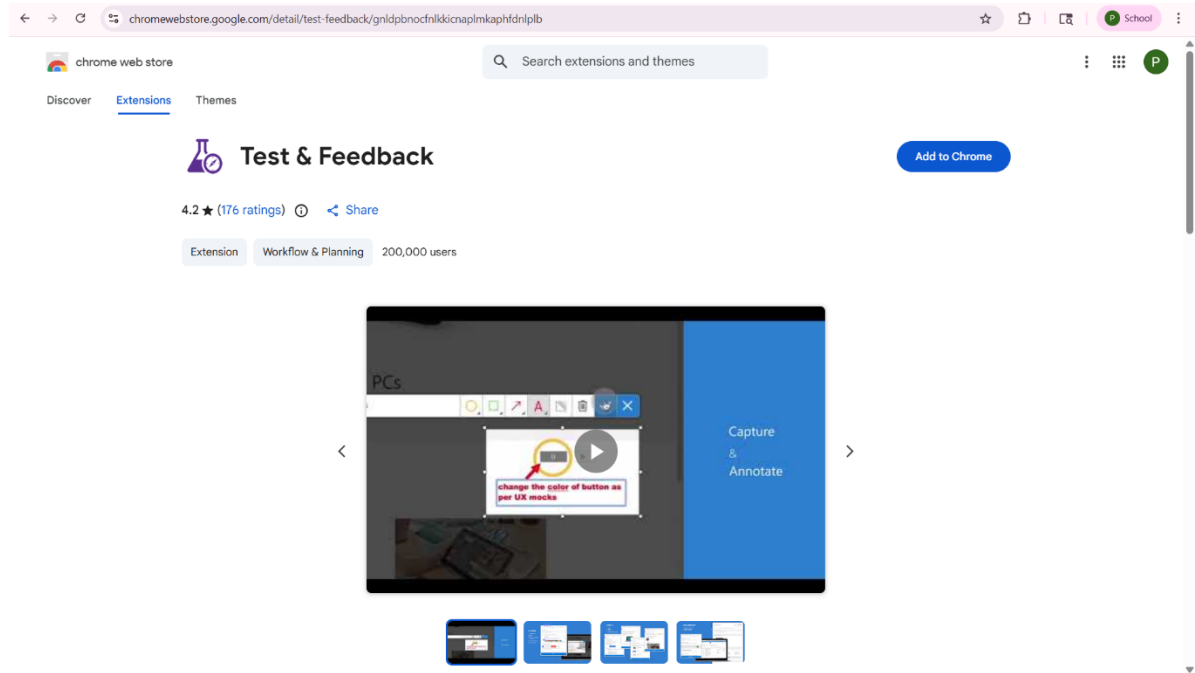
Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

[Add an existing work item as a parent](#)

4. Installation of test



5. Running the test cases

dev.azure.com/prathukshav2024aiml/Expense%20Tracker%20System/_testPlans/execute?planId=31&suiteId=33

Expense Tracker System

Expense management test cases (ID: 33)

Test Points (8 items)

Title	Outcome	Order	Test Case Id	Status
☒ TC-01 Login with valid credentials	Passed	1	34	De
☐ TC-02 Login with invalid credentials	Failed	2	35	De
☐ TC-03 Add expense with valid data	Passed	3	36	De
☐ TC-04 Add expense with empty field		37		De
☐ TC-05 Edit expense		38		De
☐ TC-06 Delete expense		39		De
☐ TC-07 View expenses	Passed	7	40	De
☐ TC-08 Generate report	Passed	8	41	De

Runner - Test Plans - Google Chrome

dev.azure.com/prathukshav2024aiml/Expense%20Tracker%20System/_testExecution/index

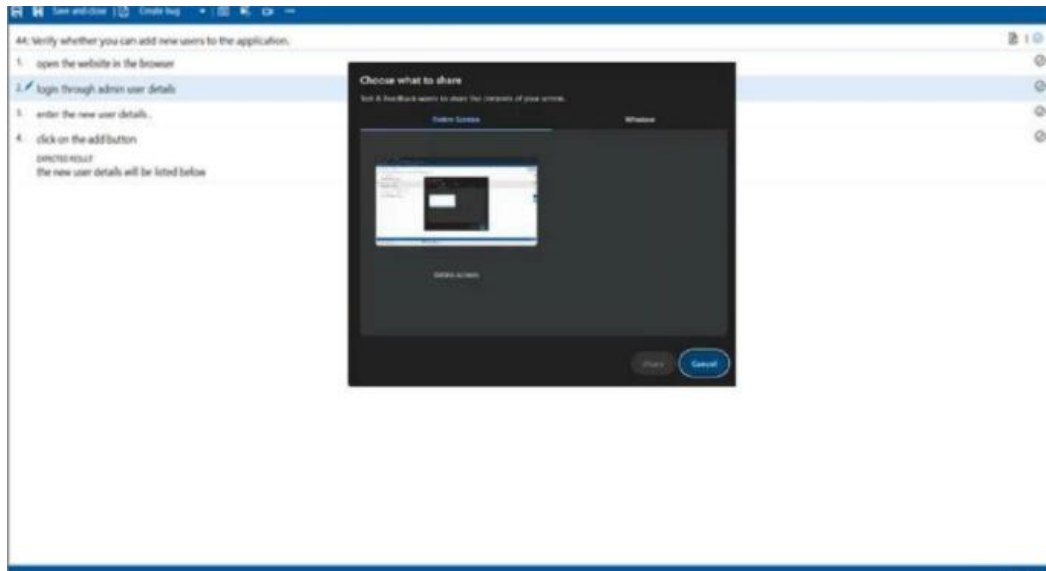
34*: TC-01 Login with valid credentials

- open applications
EXPECTED RESULT
login successful
- Enter valid username and password
- click login

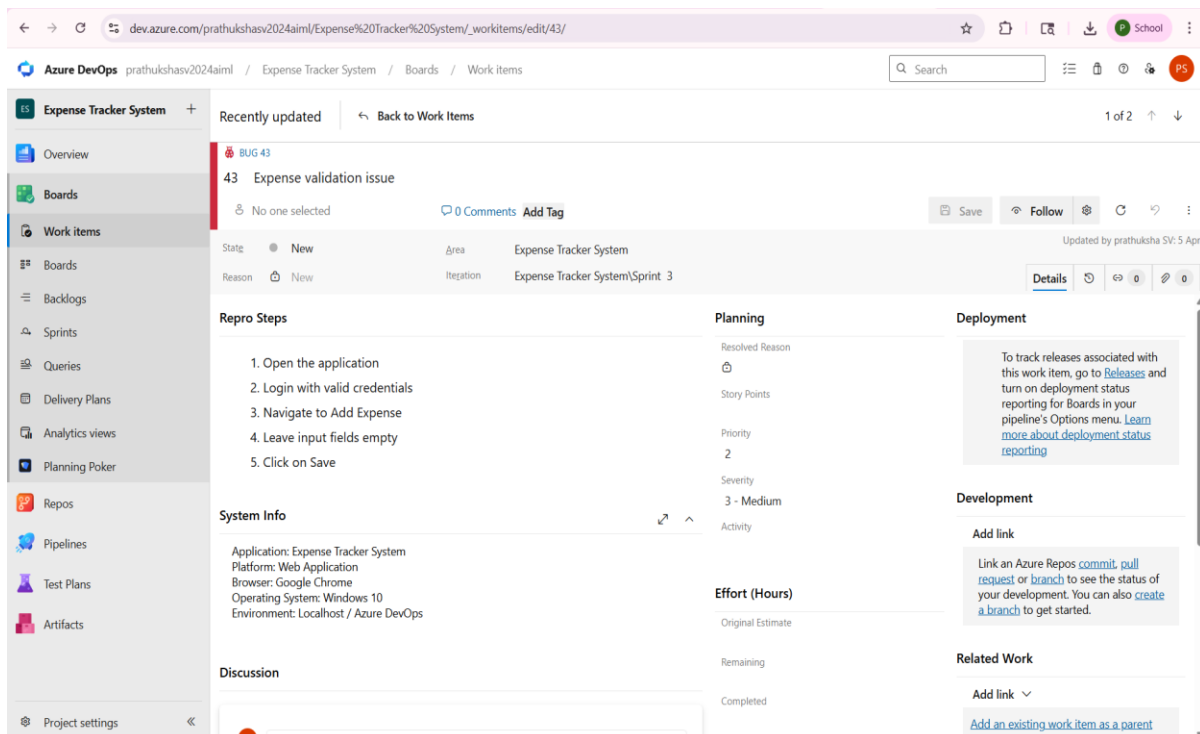
Run for web application

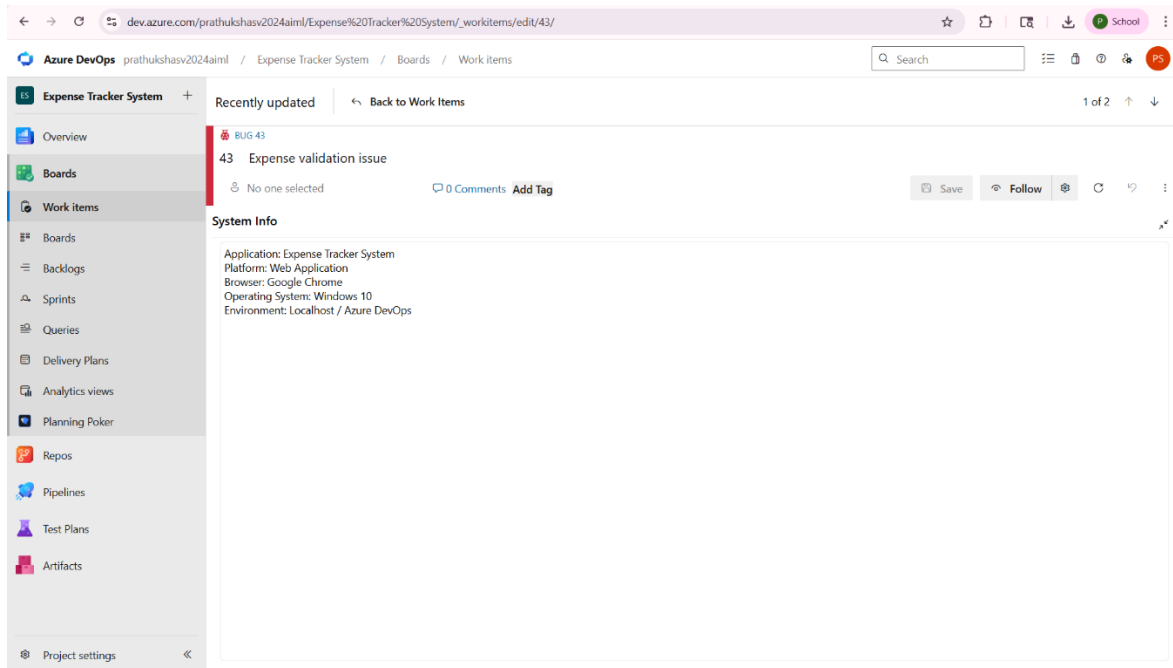
Order	Test Case Id	Status
1	34	De
2	35	De
3	36	De
4	37	De
5	38	De
6	39	De
7	40	De
8	41	De

6. Recording the test case

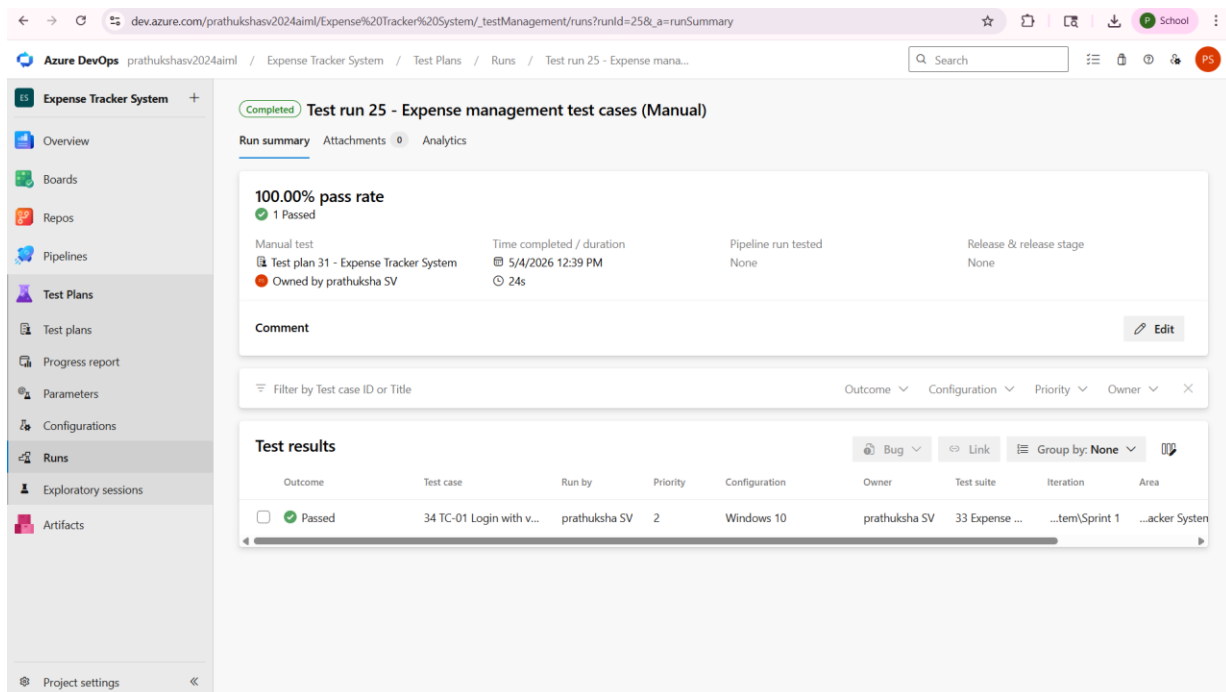


7. Creating the bug





8. Test Results



The screenshot shows the Azure DevOps Test Plans interface for the 'Expense Tracker System'. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main area displays 'Expense management test cases (16)' with a list of test points. A modal titled 'Test point ID 1 execution history' is open, showing a table of execution results.

Outcome	Test run	Run by	Time completed
Passed	25	prathuksha SV	Yesterday
Passed	2	prathuksha SV	4 Apr

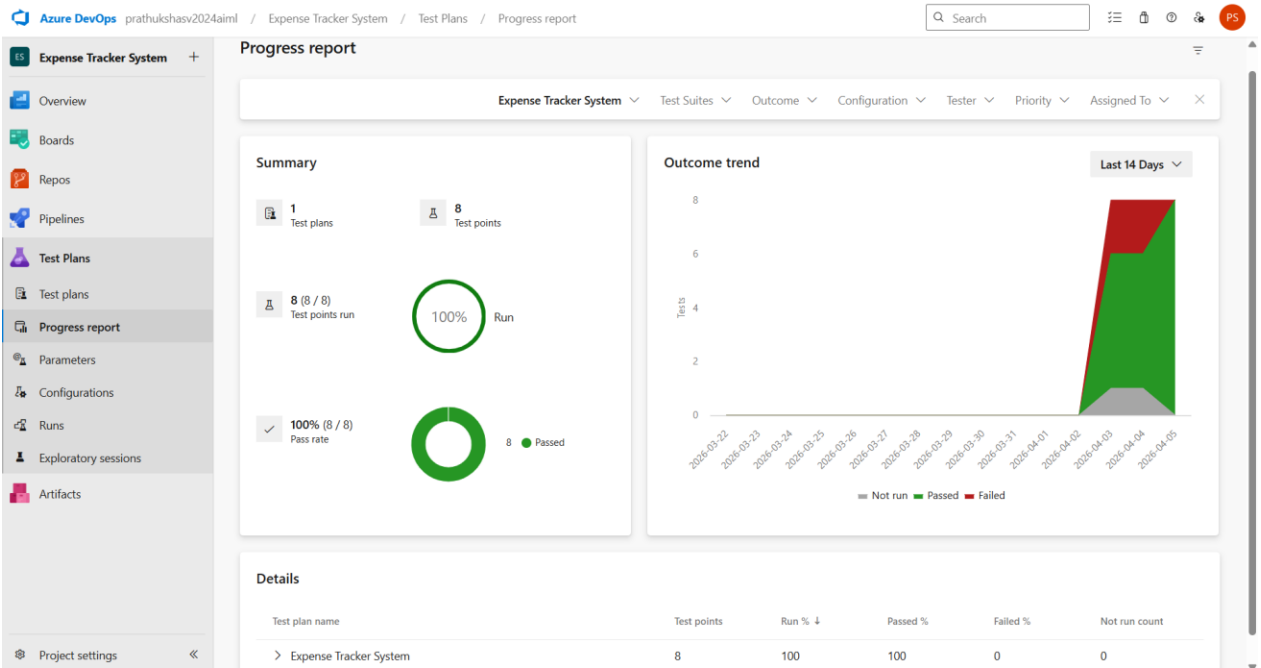
Below the table, a message states: 'Currently showing a test point execution history. View all history for a full test case execution history.' with a 'View all history' button.

8. Test report summary

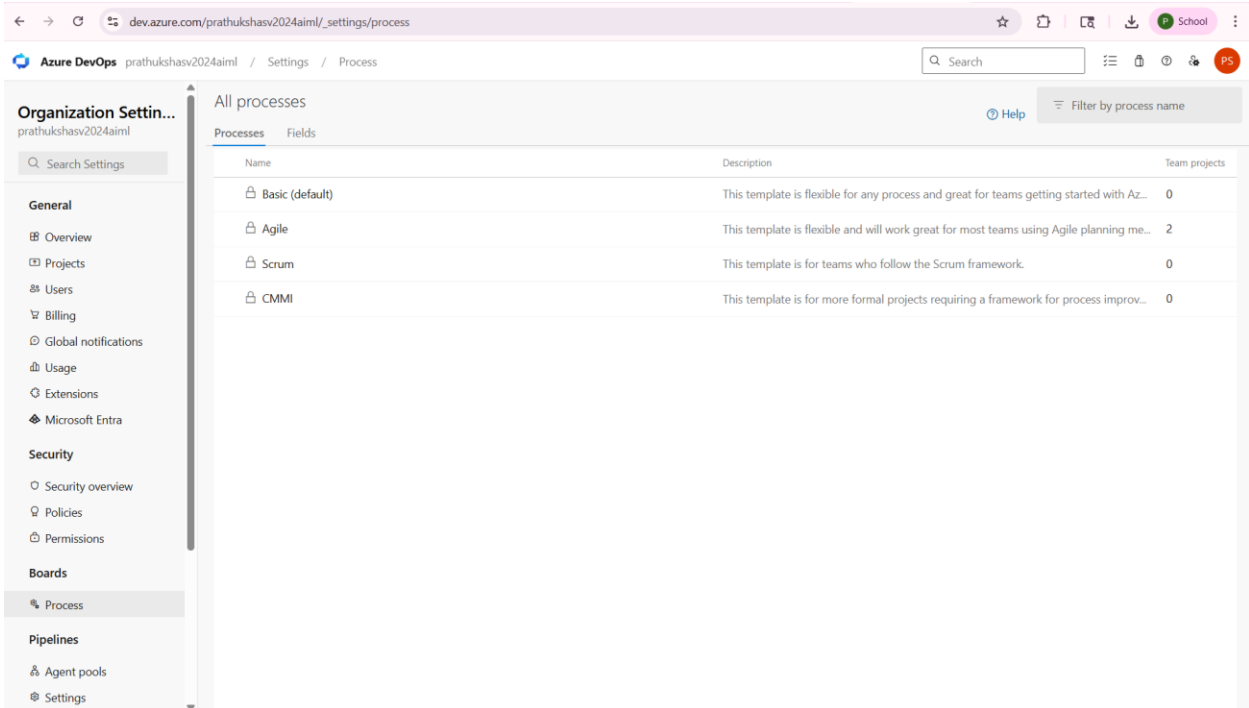
The screenshot shows the Azure DevOps Work Items interface for the 'Expense Tracker System'. The left sidebar contains navigation links: Overview, Boards, Work Items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Artifacts. The main area displays a 'BUG 43' titled 'Expense validation issue'. The 'State' dropdown is set to 'New', and the 'Reason' dropdown is set to 'New'. The 'System Info' section shows: Application: expense-tracker-system, Platform: Web Application, Browser: Google Chrome, Operating System: Windows 10, Environment: Localhost / Azure DevOps. The 'Effort (Hours)' section shows: Original Estimate, Remaining, Completed. The 'Planning Poker' section shows: Voted: 0, with buttons for 0, 1, 2, 3, 5, 8, 13. The 'Discussion' section has a comment box with the text: 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.' and a 'switch to Markdown editor' link. The 'Development' section has an 'Add link' button and a text box with the text: 'Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.' The 'Related Work' section has an 'Add link' button and a text box with the text: 'Add an existing work item as a parent'. The 'System Info' section shows: Found in Build, Integrated in Build.

1. Assigning bug to the developer and changing state

9. Progress report



10. Changing the test template



The screenshot shows the Azure DevOps interface for managing process templates. The left sidebar is titled 'Organization Settings' and includes sections for General, Security, Boards, and Pipelines. The 'Boards' section is expanded, and 'Process' is selected. The main content area, titled 'All processes', shows a table of process templates. The table has columns for Name, Description, and Team projects. The templates listed are Basic (default), Agile, Scrum, and CMMI.

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Az...	0
Agile	This template is flexible and will work great for most teams using Agile planning me...	2
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improv...	0

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9 DATE:	CI/CD PIPELINES IN AZURE
----------------------------------	---------------------------------

AIM:

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the Student Management System, ensuring faster delivery and improved software quality.

PROCEDURE:

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline
Go to the Pipelines section on the left panel and click “Create Pipeline”.
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.
If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file
6. Install dependencies (e.g., npm install, dotnet restore)
7. Build the application (dotnet build, npm run build)
8. Run unit tests (dotnet test, npm test)
9. Publish build artifacts to be used in the release stage

10. Save and Run the Pipeline for the First Time

Save the YAML or build definition and click “Run”.

Azure will fetch the latest code and execute the defined build and test stages.

11. Configure Continuous Deployment (CD)

Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.

Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals

Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.

For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs

View all pipeline runs under the Runs section.

Check logs for build/test/deploy stages to debug any errors.

You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines

Regularly update your pipeline tasks or YAML configurations as your application grows. Ensure pipeline runs are clean and artifacts are stored securely.

Integrate quality gates and code coverage policies to maintain code quality.

dev.azure.com/prathukhasv2024aiml/Expense%20Tracker%20System/_build/results?buildId=18&view=results

Expense Tracker System

Overview
Boards
Repos
Pipelines
Pipelines
Environments
Library
Test Plans
Artifacts

#20260504.1 • Set up CI with Azure Pipelines

prathukhasv.Expense-Tracker-System---software-construction-project-

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary Code Coverage

Pull request by prathukhasv

Repository and version	Time started and elapsed	Related	Tests and coverage	1 commit	Parameters
prathukhasv/Expense-Tracker-System---software-constr...	Just now	0 work items	Get started		
azure-pipelines 4bb59878	0s	0 artifacts			

Stages Jobs

Stage	
1 job completed	7s
Job	3s

Rerun Stage

Project settings

dev.azure.com/prathukhasv2024aiml/Expense%20Tracker%20System/_apps/hub/ms-vs-build-web.ci-designer-hub?pipelineId=1&nonce=v6gW2f45XyDneIS9WA%3D%3D&branch=azure-pipelines

Expense Tracker System

Overview
Boards
Repos
Pipelines
Pipelines
Environments
Library
Test Plans
Artifacts

prathukhasv.Expense-Tracker-System---software-construction-project-

Variables Run

azure-pipelines prathukhasv/Expense-Tracker-System---software-construction-project- / azure-pipelines.yml

```
1 # Starter pipeline
2 # Start with a minimal pipeline that you can customize to build and deploy your code.
3 # Add steps that build, run tests, deploy, and more:
4 # https://aka.ms/yaml
5
6 trigger:
7   - main
8
9 pool:
10  - vmImage: ubuntu-latest
11
12 steps:
13   - script: echo Hello, world!
14     displayName: 'Run a one-line script'
15
16   - script: |
17     echo Add other tasks to build, test, and deploy your project.
18     echo See https://aka.ms/yaml
19     displayName: 'Run a multi-line script'
20
```

Tasks

Search tasks

- .NET Core
- Android signing
- Ant
- App Center distribute
- App Center test
- Archive files
- ARM template deployment
- Azure App Configuration Export
- Azure App Configuration Import
- Azure App Configuration Snapshot
- Azure App Service deploy

Project settings

RESULT:

Thus the pipelines for the given project “Expense tracker System” has been executed successfully

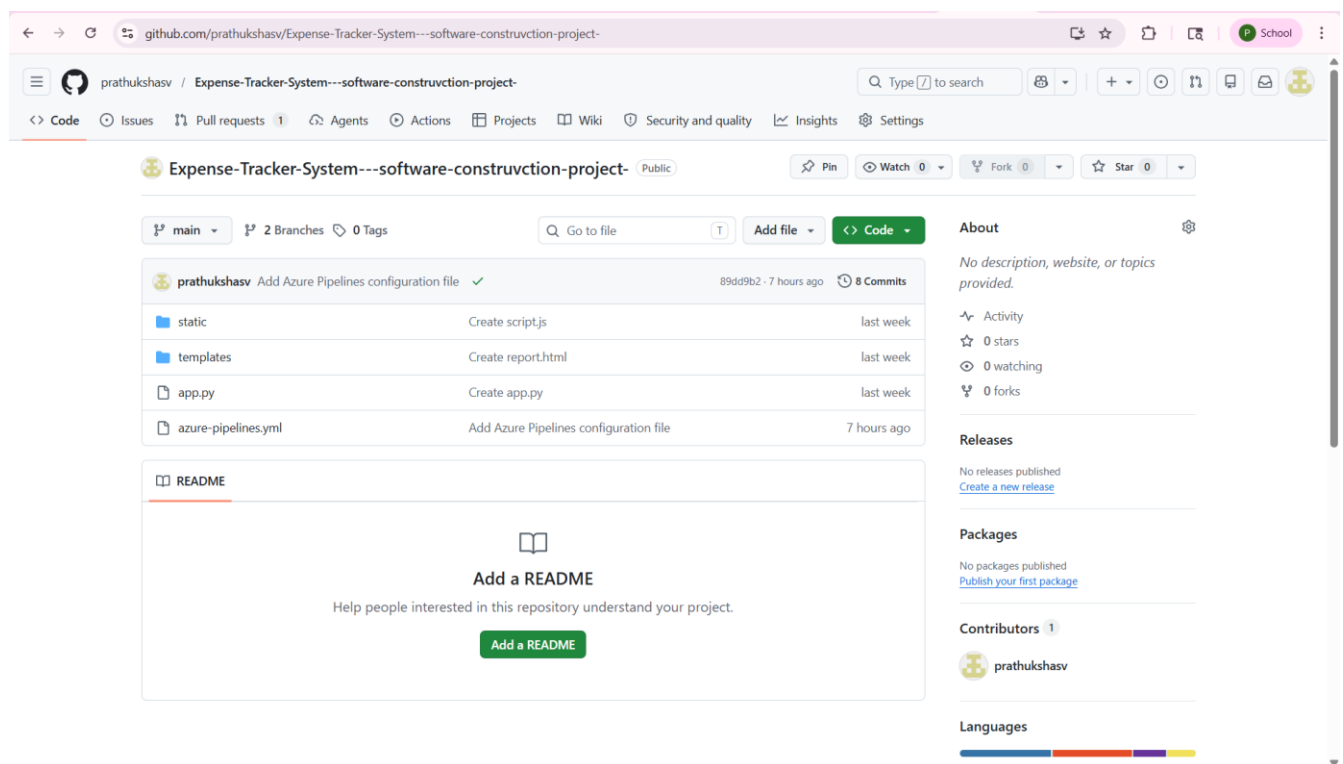
EXP NO: 10
DATE:

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Student Management System project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase